

Disambiguating “Perimeter” Across the Trilogy

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This work formalizes a cross-paper conceptual disambiguation across the three papers of the Coordination Knowledge Substrate (CKS) theory series: “Coordination Outside the Model” (Li, April 2026), “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026).

It does not introduce new axioms. Its sole contribution is to articulate, in operational form, the precise meaning of three related but distinct perimeter concepts introduced across the trilogy — home perimeter, inter-Self perimeter, and the additive perimeter property — so that downstream work can adopt or argue against these terms without ambiguity.

Abstract

The CKS trilogy introduces perimeter concepts progressively. Paper 1 implies a governance boundary through its outside-the-model commitment but does not name a perimeter explicitly. Paper 2 names the **home perimeter**: the bounded governance architecture of a complete Self, encompassing all three tiers and both layers, governed by that Self's home authority. Paper 3 adds two further concepts: the **inter-Self perimeter**, the governance boundary that Full Aspect Integration (FAI) crosses when two or more Selves coordinate through a shared substrate; and the **additive perimeter property**, the architectural commitment that participation in FAI adds an inter-Self governance scope without reducing or replacing the home perimeter. This note disambiguates all three, identifies the adversarial misreading each is designed to prevent, and provides a coexistence test for compliant FAI. The additive perimeter property receives the most extended treatment because it is the most consequential disambiguation: any reading that FAI erodes home governance misunderstands the architecture at its foundation.

1. Why disambiguation is needed

The word “perimeter” appears across Papers 2 and 3 with different referents at different scopes, and the progression across papers makes it easy to import an earlier referent into a later context where a different — or additional — concept is intended.

Three failure modes motivate this note.

Failure mode 1 — Collapsing home and inter-Self perimeter. A reader who treats “perimeter” uniformly across Paper 3 may assume that every occurrence refers to the home perimeter, missing

the distinct governance structure the inter-Self perimeter names. This misreading produces a flattened account of Paper 3 in which FAI appears to operate within each Self's existing home governance architecture, rather than spanning a new cross-organizational boundary with its own governance logic.

Failure mode 2 — Reading FAI as perimeter replacement. A reader who notices that FAI introduces a new perimeter concept may conclude that the inter-Self perimeter supersedes or replaces the home perimeter during FAI participation. This misreading is the most consequential: it implies that a Self's home governance authority is suspended or diluted during FAI, which contradicts a foundational architectural commitment of the trilogy. The additive perimeter property is the direct response to this misreading.

Failure mode 3 — Treating Paper 1's implicit boundary as unrelated. A reader who does not trace the genealogy of “perimeter” concepts may treat Paper 2's home perimeter as an independently introduced concept with no grounding in Paper 1. This misreading severs the continuity of the governance boundary concept across the trilogy, making Paper 2's naming appear ad hoc rather than as the explicit formalization of what Paper 1 already commits to implicitly.

Each of the three perimeter concepts this note disambiguates is the response to one of these failure modes.

2. Paper 1's implicit substrate boundary: the historical basis

Paper 1 does not use the word “perimeter” as an explicit concept. What Paper 1 commits to instead is the **outside-the-model spine**: the architectural commitment that coordination knowledge, governance semantics, and conflict-handling logic live in a persistent substrate outside the LLM, not inside LLM inference and not in specialized runtime middleware. This commitment implies a boundary — there is a governance space (the human-governed substrate) and an execution space (the LLM operating under orchestration rules) — but Paper 1 does not name that boundary “perimeter.”

The substrate's governance boundary in Paper 1 is defined by the human-governed commitment: within the substrate, humans retain the right to inspect, modify, and override content and orchestration rules at any time. Outside the substrate, the LLM executes under rules but does not hold governance authority. This inside-versus-outside structure is the historical basis for what Paper 2 will explicitly call a perimeter.

The practical consequence for downstream readers is that the home perimeter Paper 2 names is not a new architectural commitment; it is the explicit formalization of a boundary that Paper 1's six commitments jointly require. Prior-art readers who encounter “home perimeter” in Paper 2 should read it as the named version of Paper 1's already-committed governance boundary, now made explicit because Paper 2's three-level structure (cell, aspect, Self) requires naming the outermost boundary precisely.

3. Home perimeter (Papers 2 and 3): the bounded governance architecture of a Self

Paper 2 introduces the **home perimeter** as the outermost governance boundary of a complete Self's architecture. The home perimeter:

- Encompasses all three tiers of one Self's governance architecture: cells, aspects, and the Self level.
- Covers both layers within every cell and aspect: the DNA layer (orchestration substrates, behavior substrates, schemas, rules) and the action layer (recorded task instances, outputs, operational evidence).
- Is bounded by that Self's home governance authority — the humans who hold the three rights (inspect, modify, override) over this Self's substrate content and orchestration rules.
- Does not automatically extend to other Selves' architectures. A second Self's governance authority is not home authority for the first Self's substrate; the two are architecturally separate.

At Paper 2 scope, the home perimeter is the only perimeter concept in the architecture. All operations — lifecycle events, evolution mechanism outputs, combination primitives, conflict handling — take place within one Self's home perimeter. There is no cross-perimeter operation at Paper 2 scope; the architecture that would enable such operations is Paper 3's territory.

Paper 3 inherits the home perimeter concept unchanged. During FAI participation, a Self's home perimeter remains intact — it does not dissolve, contract, or merge with other Selves' home perimeters. Paper 3 explicitly names this as **home perimeter integrity** (the verification that each participating Self's home governance architecture is undisturbed by the FAI event). The home perimeter is the stable boundary within which home governance authority operates throughout FAI, before it, and after it.

4. Inter-Self perimeter (Paper 3): the cross-organizational boundary FAI crosses

Paper 3 introduces the **inter-Self perimeter** as the governance boundary between different Selves' home governance architectures. The inter-Self perimeter is not a property of any single Self; it is a relational boundary that exists at the interface between two or more Selves' home perimeters when those Selves coordinate through a shared substrate.

The inter-Self perimeter has a different governance structure from the home perimeter.

What governs the crossing. Exchange bounding governs what content may cross the inter-Self perimeter. The exchange is bounded to substrate content — specifically, DNA-layer content (orchestration substrates, behavior substrates, schemas, rules) and action-layer content (recorded task instances, outputs, operational evidence). LLM weights and instinct-layer content do not cross the inter-Self perimeter. This bounding is an architectural commitment, not a configurable parameter: it is the extension of Paper 2's instinct/reasoning separation to the inter-Self scope.

Who governs what crosses. Joint authority governs the content that enters the shared substrate after crossing the inter-Self perimeter. Both participating Selves' governance structures hold authority over the shared substrate's content and orchestration rules. Neither Self's home authority alone is sufficient to govern the shared substrate; governance of what crosses is jointly held.

What the crossing produces. FAI — Full Aspect Integration — is the canonical operation at the inter-Self scope. When aspects from two or more Selves exchange content through the shared substrate, that exchange takes place across the inter-Self perimeter. The shared substrate is temporarily constructed for the FAI event, carries all six Paper 1 commitments within its scope, and dissolves at FAI completion.

The inter-Self perimeter is not present at Paper 2 scope. It comes into existence when a shared substrate is constructed for an FAI event and ceases to be operationally active when that event dissolves. It is a Paper 3 concept with no Paper 2 antecedent beyond the implication that home perimeters are bounded and therefore have edges that would be crossed by cross-organizational coordination.

5. The additive perimeter property (Paper 3): the governance sovereignty protection

The **additive perimeter property** is the architectural commitment that FAI participation adds an inter-Self governance scope to each participating Self's existing home perimeter — it does not replace, reduce, or suspend the home perimeter.

This property is the most important disambiguation in this note, for the following reason: the misreading it forecloses is the most consequential. If a reader interprets FAI as replacing the home perimeter with a joint-authority structure for the duration of the event, the conclusion follows that each Self's home governance authority is partially surrendered during FAI. This conclusion is architecturally incorrect, and its incorrectness is load-bearing: the trilogy's claim that multi-organizational AI coordination is compatible with each organization maintaining full governance sovereignty over its own architecture depends entirely on the additive property holding.

The additive perimeter property commits to the following:

1. During FAI, the home perimeter remains fully intact. Each participating Self's home governance authority continues to operate over that Self's home substrate content without interruption. No content in a Self's home substrate becomes subject to the joint authority that governs the shared substrate; joint authority applies only to the shared substrate, not to either Self's home substrate.

2. The inter-Self perimeter is added, not substituted. The shared substrate's inter-Self governance scope is an additional governance layer present during the FAI event, not a replacement for either Self's home governance layer. The participating Selves operate under two governance scopes simultaneously: their home governance (over their home substrate content) and joint governance (over the shared substrate content).

3. Dissolution restores the single-perimeter state. When the FAI event dissolves, the inter-Self perimeter ceases to be active. Each Self returns to operating under its home perimeter only. Content that was absorbed into home substrate during dissolution is immediately subject to home governance authority; the joint authority that governed it in the shared substrate does not follow it across the perimeter.

The additive property is what makes the architecture's governance sovereignty claim coherent at any scale. At population scope — where many Selves participate in multiple FAI events over time — each Self continues to hold full home governance authority over its own substrate throughout. The accumulation of FAI events does not gradually erode home perimeters; each event adds a temporary inter-Self perimeter that dissolves completely, leaving home perimeters intact.

6. Three perimeter concepts coexist during FAI

During an active FAI event, all three perimeter concepts apply simultaneously. This coexistence is not a complication; it is the architecture.

Home perimeter (each Self): intact, governed by home authority, carrying home substrate content that does not enter the shared substrate.

Inter-Self perimeter (relational, between Selves): active for the duration of the FAI event, governing what crosses (exchange bounding) and who governs what crosses (joint authority).

Additive perimeter property (architectural): the structural fact that the inter-Self perimeter's presence does not reduce the home perimeter — both operate simultaneously without competition.

An operational test for compliant FAI follows from this coexistence requirement. A FAI operation is architecturally compliant if and only if:

- (1) Each participating Self's home governance authority remains exercisable over that Self's home substrate content throughout the FAI event — no content in any Self's home substrate has become subject to joint authority.
- (2) The shared substrate's content is subject to joint authority of all participating Selves' governance structures — no single Self's home authority governs the shared substrate unilaterally.
- (3) Content crossing the inter-Self perimeter is bounded to DNA-layer and action-layer substrate content — no LLM weights or instinct-layer content crosses.
- (4) At dissolution, content absorbed into each Self's home substrate immediately becomes subject to that Self's home governance authority — joint authority does not follow absorbed content across the perimeter.
- (5) The inter-Self governance scope is traceable as an additive layer on top of each Self's home governance scope — not as a modification of it.

A coordination mechanism that fails condition (1) has violated home perimeter integrity. One that fails condition (2) has collapsed joint governance into unilateral home governance. One that fails condition (3) has violated exchange bounding. One that fails condition (4) has allowed joint authority to persist across dissolution. One that fails condition (5) has instantiated perimeter replacement rather than the additive perimeter property.

7. Summary and prior-art disambiguation claim

Concept	Introduced	Scope	Governed by	Relation to others
Implicit substrate boundary	Paper 1 (implicit)	Cell/substrate	Home authority (implied)	Basis for Paper 2's explicit naming
Home perimeter	Paper 2 (explicit)	Complete Self (all tiers, both layers)	That Self's home authority	Remains intact during FAI (Paper 3)
Inter-Self perimeter	Paper 3	Between Selves' home architectures	Exchange bounding + joint authority	Added to home perimeter (additive property)
Additive perimeter property	Paper 3	Architectural (during FAI)	Structural commitment	Ensures inter-Self does not replace home perimeter

Prior-art disambiguation claim. The trilogy introduces perimeter concepts progressively across three papers. Paper 1's outside-the-model commitment implies a governance boundary without naming it. Paper 2 names the home perimeter explicitly as the outermost boundary of a Self's complete governance architecture. Paper 3 adds the inter-Self perimeter as the cross-organizational governance boundary FAI crosses, and the additive perimeter property as the structural commitment that inter-Self participation does not erode home governance sovereignty. These three concepts coexist at Paper 3 scope, serve distinct functions, and are not in conflict. Any reading that treats them as equivalent, or that treats FAI as reducing the home perimeter, mischaracterizes the architecture.

Source papers

Li, W. (2026a). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026b). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026c). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

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